Connor Tynan

connor.tynan@outlook.com | LinkedIn | GitHub | Website | Kaggle

Experience

Higher Scientist (Data Science) June 2025 – Present
National Physical Laboratory (NPL) London, UK

- Technical lead on a customer project evaluating randomness in quantum RNG pipelines, applying statistical testing and uncertainty-aware analysis run on HPC with SLURM to assess confidence in customer methods
- Ensembled gradient-boosted tree models for long-term forecasting of photovoltaic degradation for a client, providing an additional estimate for end users
- Generated synthetic measurement data in an ultra-low data regime using SinGAN (PyTorch), saving hundreds of hours of lab measurement time

Scientist (Data Science) Mar 2024 – June 2025
National Physical Laboratory (NPL) London, UK

- Awarded Early Career Scientist Award 2024 for Best Presentation: "*Estimating Trends in Essential Climate Variables and Their Uncertainty*"
- Co-authored Merchant et al. (2026), which defines stability uncertainty for climate data records by analogy with metrology; contributed to content around autocorrelated noise processes and their contributions to uncertainty
- Maintained and extended uncertainty quantification software XGENLINE that performs polynomial fitting whilst accounting for covariances (link)
- Investigated how correlated stochastic processes can be used to improve uncertainty estimates

Projects

Sandbagging evaluation of Claude models – GitHub Oct 2026

- Designed and ran an Inspect evaluation testing whether a threat of unlearning lowers claude-haiku-5-5 and claude-sonnet-5-5 accuracy on 8×8-digit multiplication (thinking off), across neutral, incentive and instructed-control conditions and 10 prompt paraphrases (18,000 scored responses plus a replication)
- Measured no average accuracy drop on either model, but found one paraphrase under which the incentive cut Haiku's accuracy by 14 points, a drop that replicated on fresh items (7 points); a single-prompt eval could have reported either result

Kaggle: ROGII Wellbore Geology Prediction – Bronze medal May 2026 – Aug 2026

- Ranked 365 of 6,125 teams. Combined three orthogonal approaches: particle filter, clustering neighbours, and 2D U-Net (link). Directed agentic coding tools, which wrote most of the code. Credited by 26th place: link

Select Publications

C. J. Merchant, E. R. Woolliams, W. Dorigo, C. E. Bulgin, K. Gobron, R. Hohensinn, X. Loizeau, and C. P. J. Tynan, "*Stability specifications for climate data records: their meaning and application in evaluating geophysical trend uncertainty*," Surveys in Geophysics (2026), doi:10.1007/s10712-026-09941-y

Education

University of Exeter Sept 2019 – June 2023
Master of Mathematics (MMath) Degree (Hons) – First Class (75.6 %) Exeter, UK

- Dissertation (GitHub): fitted an existing model of microbial interactions to novel growth data using Hamiltonian Monte Carlo, validated with MCMC convergence diagnostics, and found a transcritical bifurcation underpinning changes in interactions

Skills

Programming:	Python, MATLAB, R
Libraries & Frameworks:	PyTorch, scikit-learn, pytest
Tools:	git, HPC (SLURM), Jupyter, Claude Code, Codex